

Decent Coaching Classes

VIII CBSE

Mind Your Math - IV

Answer Key — Squares & Cubes, Powers & Exponents, Quadrilaterals, Playing with Numbers

Class VIII — Mathematics (NCERT) — Ch 1, 2, 4 & 5 | Maximum Marks: 30

Section A — Multiple Choice Questions

- A1.** (b) 6 — since the units digit of 7014 is 4, and $4^2 = 16$, the units digit of the square is 6. [1]
- A2.** (a) 2401 — since $49^2 = 2401$, it is a perfect square (2402, 2403, 2404 are not, as no whole number squares to any of them). [1]
- A3.** (d) -128 — $(-4)^2 = 16$ and $(-2)^3 = -8$, so $16 \times (-8) = -128$. [1]
- A4.** (b) a^3b^2 — three a's multiply to a^3 and two b's multiply to b^2 . [1]
- A5.** (b) 70° — adjacent angles of a parallelogram are supplementary, so $180^\circ - 110^\circ = 70^\circ$. [1]
- A6.** (b) 753 — digit sum = $7 + 5 + 3 = 15$, which is divisible by 3 (sums for the other options — 10, 23, 19 — are not). [1]

Section B — Fill in the Blanks / Very Short Answer

- A7.** 18 (since $324 = 2^2 \times 3^2 \times 3^2 = 2^2 \times 3^4$, so $\sqrt{324} = 2 \times 3^2 = 18$). [1]
- A8.** $1/16$ [1]
- A9.** equal [1]
- A10.** 4 [1]

Section C — Short Answer Questions

- A11.** Using the long division method, $\sqrt{1225} = 35$. (Grouping as 12'25, the largest number whose square is ≤ 12 is 3 ($3^2 = 9$, remainder 3); bringing down 25 gives 325; doubling the quotient 3 gives 6, and the digit d such that $6d \times d \leq 325$ is $d = 5$, since $65 \times 5 = 325$, remainder 0. So the quotient is 35.) [2]
- A12.** $392 = 2^3 \times 7^2$. Since the power of 7 is 2 (not a multiple of 3), 392 is not a perfect cube. Dividing by $7^2 = 49$ removes the extra factor of 7, leaving $392 \div 49 = 8 = 2^3$, which is a perfect cube. So the smallest natural number to divide by is 49. [2]
- A13.** $15^{15} = (3 \times 5)^{15} = 3^{15} \times 5^{15}$. So $15^{15} \div (3^{12} \times 5^{12}) = 3^{15-12} \times 5^{15-12} = 3^3 \times 5^3 = (3 \times 5)^3 = 15^3$. [2]
- A14.** Sum of angles of a quadrilateral = 360° . So $5x + (2x + 20) + (3x + 10) + 70 = 360 \rightarrow 10x + 100 = 360 \rightarrow 10x = 260 \rightarrow x = 26$. The four angles are: $5(26) = 130^\circ$, $2(26) + 20 = 72^\circ$, $3(26) + 10 = 88^\circ$, and 70° . (Check: $130 + 72 + 88 + 70 = 360^\circ$.) [2]
- A15.** Adjacent angles of a parallelogram are supplementary and opposite angles are equal. So if one angle is 120° , the angle adjacent to it is $180^\circ - 120^\circ = 60^\circ$, and since opposite angles are equal, the four angles are $120^\circ, 60^\circ, 120^\circ$, and 60° . [2]

A16. $37zy$ has digit sum $3 + 7 + z + y = 10 + z + y$. For $37zy$ to be a multiple of 3, this sum must be a multiple of 3, i.e., $10 + z + y \equiv 0 \pmod{3}$, which gives $z + y \equiv 2 \pmod{3}$. Since z and y are digits (0–9 each), $z + y$ can range from 0 to 18, so the possible values of $(z + y)$ are 2, 5, 8, 11, 14 and 17. [2]

Section D — Long Answer Questions

A17. Since $ABCD$ is a rectangle, its diagonals are equal and bisect each other, so $OA = OB = OC = OD$. In triangle OAB , since $OA = OB$, it is isosceles with base angles $\angle OAB = \angle OBA = 34^\circ$, so (ii) $\angle ABO = 34^\circ$. Then $\angle AOB = 180^\circ - 34^\circ - 34^\circ = 112^\circ$, and since A, O, C are collinear, $\angle BOC = 180^\circ - \angle AOB$, so (iv) $\angle BOC = 180^\circ - 112^\circ = 68^\circ$. Also $\angle COD$ is vertically opposite to $\angle AOB$, so (iii) $\angle COD = 112^\circ$. Finally, in triangle ABC , $\angle ABC = 90^\circ$ (angle of the rectangle) and $\angle BAC = \angle OAB = 34^\circ$, so (i) $\angle ACB = 180^\circ - 90^\circ - 34^\circ = 56^\circ$. [4]

A18. Starting at 37 and adding 14 repeatedly gives the sequence: 37, 51, 65, 79, 93, 107. Their digital roots are: $37 \rightarrow 3+7=10 \rightarrow 1$; $51 \rightarrow 5+1=6$; $65 \rightarrow 6+5=11 \rightarrow 1+1=2$; $79 \rightarrow 7+9=16 \rightarrow 1+6=7$; $93 \rightarrow 9+3=12 \rightarrow 1+2=3$; $107 \rightarrow 1+0+7=8$. So the digital roots are 1, 6, 2, 7, 3, 8. The pattern observed is that the digital root increases by 5 each time, and whenever it would exceed 9, it wraps around by subtracting 9 (i.e., it alternately increases by 5 and effectively decreases by 4, since adding 14 changes the digital root by $14 \bmod 9 = 5$). [4]

— End of Answer Key —